Biochemistry Laboratory Course for BDS Students

PRACTICAL JOURNAL

INDEX

	Date	Topics	Content	Page No	Signature /Remarks
1		**Introduction to Lab Safety Procedures** • To be aware of Lab safety procedures.	1. Lab Safety Procedures 2. Principles and Operating Procedures of Lab Precautionary Measures 3. Lab Safety & Hazards 4. Videos: Burns (Chemical, Fire), Chemical Handling, Biohazards, Disposal of Chemical Wastes		
2		**Introduction to Lab Equipment** • Principle and operating procedures of lab equipment.	1. Thermostatic/Water Bath 2. Shaker/Mixer 3. Spectrophotometer 4. Electrical/Analytical Balances 5. Oven 6. Glucometer 7. pH Meter 8. Chromatography 9. Electrophoresis 10. Autoanalyzer in Biochemistry Lab		
3		**Observation of the common practice of Sample drawing Techniques** • Visit to the biochemistry working lab. • Videos for instrument use.	1. Venipuncture 2. ABGs 3. CSF Collection 4. Capillary Sampling		

	Date	Topics	Content	Page No	Signature /Remarks
4		**Thorough Overview of samples and Sample collection techniques** • Types of Samples • Types of collection methods • Types of vacutainers and their color coding	1. Whole Blood 2. Serum 3. Plasma 4. Types of anticoagulants and their applications 5. Urine 6. CSF		
5		**Biochemical Sample Use for Analysis and Its Interpretation**	1. Blood 2. Serum 3. Plasma 4. ABGs 5. Urine sampling		
6		**Preparation of Biochemical Solutions** • Define solution, its types. • Preparation of solutions of different concentrations	1. Normal Solution, Molar, Molal, Osmolar 2. Preparation of Buffers 3. Working and Stock Solutions		
7		**Effective Sample Processing Techniques** • Visit to Biochemistry Working Lab • Videos for Instrument and process	**Basic Equipment and Techniques:** 1. Centrifuge 2. Spectroscopy 3. Electrochemistry 4. Chromatography 5. Electrophoresis 6. Enzymology 7. Immunological Tests (Hepatitis, PCR, Pregnancy **8.** Use of Autoanalyzer		
			Advanced Biochemical		

	Date	Topics	Content	Page No	Signature /Remarks
			Techniques 1. Protein Purification 2. Homogenization 3. LDH Purification 4. Enzyme Assays 5. Gel Filtration 6. Protein Crystallography 7. ELISA		
			Molecular Biology Techniques: 1. Northern Blotting (RNA) 2. Southern Blotting (DNA) 3. Western Blotting (Protein)		
8		**Manual Protocols**	**CHO Detection Methods:** 1. Molish" s Test 2. Iodine Test 3. Benedict Test 4. Barford Test 5. Thymol Blue Test 6. Seliwaoff's Test 7. Osazoe Test Carbohydrate-Related Disorders (Diabetes Mellitus)		
		Carbohydrate Detection • Define the principle and procedure for CHO detection methods. • Identify and differentiate sugars-non-sugars, reducing-non-reducing sugars, and monosaccharide-polysaccharides in any sample/solution.			
9		**Identification of Fats and Lipids** • Define the hydrophobic nature of fats. • To identify hydrophobic and	1. Hydrophobic Nature of Fats 2. Emulsification Test: Greasy Nature of Lipid 3. Rancidity Detection 4. Cholesterol Detection (Salkowski Test) 5. Saponification, Acrolein Test		

	Date	Topics	Content	Page No	Signature /Remarks
		hydrophilic solutions.	6. Estimation of Total Cholesterol (Enzymatic Colorimetric CHOD-PAP) 7. Disorders like Atherosclerosis		
10		**Advanced Protein Detection Methods** • To detect the protein in any sample/solution.	1. Tests: Biuret Test 2. Lead Acetate Test 3. Sulphosalicylic Acid Test 4. Xanthoproteic Test 5. Hopkins-Cole Test 6. Millon's Test 7. Lead Sulfide Test 8. Ninhydrin Test 9. Acetic Acid Test 10. Neumann's Test (Confirmatory Test for Casein) 11. Heat Coagulation Test 12. Salt Saturation Tests Disorders like Kwashiorkor, Marasmus, Renal Disorders, and Cirrhosis		
11		**Systematic Amino Acid Identification and Separation** • Define Polar and Non-Polar Amino Acids	1. Chromatography 2. Centrifugation 3. Salting Out		
12		**Comprehensive Protein Separation** • To describe Ionic character of proteins. • Define principle and procedure of electrophoresis	1. Ionic Character of Proteins 2. Principle and Procedure of Electrophoresis		
13		**Enzyme Activity: Temperature**	1. Enzyme Activity 2. Effect of Temperature and		

	Date	Topics	Content	Page No	Signature /Remarks
		• Define Enzyme activity. • Discuss the effect of temperature and pH on enzyme activity.	Ph on Enzyme Activity		
14		**Detailed Analysis of Macromolecules**			
		Bile salt & Bile pigments • To detect the bile salts and bile pigments in given solution.	**1.**Detection of Bile Salts and Bile Pigments in Solution		
		Normal and Abnormal Urine Constituents • Enlist the abnormal contents of urine.	1. Correlate the abnormal constituents of urine with the clinical condition.		
		Determination of Uric Acid • Define Normal value of uric acid.	1. Correlate the abnormal value of uric acid with the disease.		
15		**Interpretation of Hemoglobin Estimation &Liver Function Test (LFT)**	1. Hemoglobin estimation by enzymatic method 2. Liver Enzyme Estimation Principles and Procedures. 3. Normal and Abnormal Values of Liver Enzymes. 4. Abnormalities of Bile Pigments and Liver Enzymes in Relation to Jaundice and Other Abnormalities		
16		**Analysis of milk**	1. Composition of milk analysis(Casein) sample milk 2. Newman's test		

	Date	Topics	Content	Page No	Signature /Remarks
			3. Color test		
17		**Thyroid Profile**	1. TSH, T3 &T4 importance with clinical conditions of the thyroid glands 2. Thyroid hormone estimation		

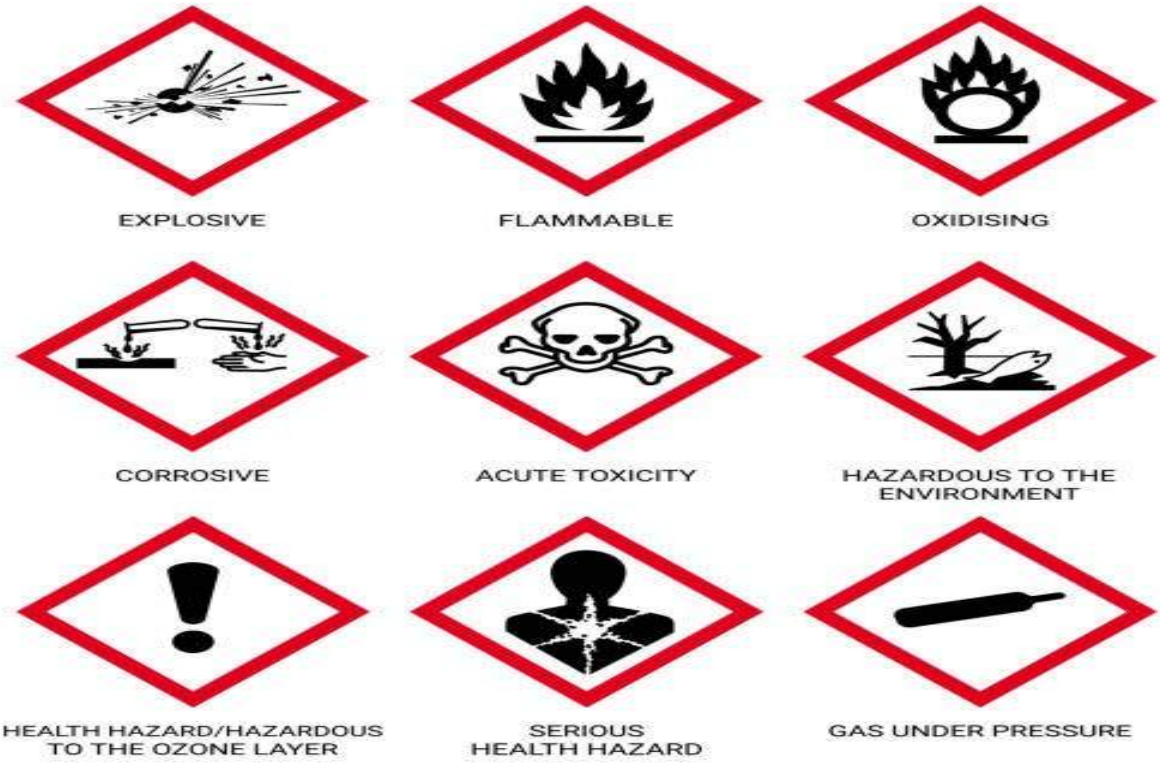
Chapter 1: to Lab Safety Procedures

Introduction to Lab Safety Procedures

laboratory safety rules:

The laboratory safety rules are considered as a starting point for safe laboratory practices. The following comments on the general rules listed may help to clarify any questions which arise.

1. Working without supervision is always forbidden.
2. Performing unauthorized experiments or any experiment at unauthorized time is also forbidden.
3. Loose hair and clothing must be restrained. Clothing worn should be capable of providing a barrier between your skin and the chemicals.
4 Eating, drinking, smoking and applying cosmetics are forbidden.
5. All accidents must be reported to an instructor.
6. During practical, pipetting by mouth is also forbidden.
7. When needed, gloves must be worn.
8. You must know the location and proper use of all laboratory safety equipment, including eyewash, safety shower, fire extinguishers, and telephone.
9. Must know the emergency exit of laboratory.
10. Items such as book bags, backpacks, purses and mobile phones should be put out of the way off the floor and on a lab bench.
11. You should have your complete stationery including a lab manual to note down observations/results of the experiment.
12. Cell phones should be turned off before entering the laboratory.
13. Read labels and dispose of any waste in an appropriate waste container.
14. Make sure you wash and put away all the equipment before you leave the laboratory.
15. Inform your instructor at the beginning of the term if you have any special medical conditions that may need attention during the laboratory, such as known allergies or seizures.



Precautionary Measures

The Four *Principles* of Safety

Recognize Hazards
Assess Risks
Minimize Risks
Prepare for Emergencies

These four principles appear in nearly every section of this book. Memorizing, and using, these ideas, whenever you think about safety issues will lead to "incident-free" laboratories

CHEMICAL SAFETY

Commonly used Signs and Symbols in labs

1. The vapors of many organic solvents are flammable or combustible. Do not expose electric sparks, open flames and heating elements to organic solvent vapors.
2. Many chemicals are poisonous. Do not taste and pipette by mouth.

3. Be extremely careful when transferring, distilling or refluxing volatile liquids. 4. Do not return used chemicals back to the stock container.
5. Do not tap flasks under vacuum.
6. Do not heat, measure or mix any chemicals in front of your face.
7 Never heat a closed system it will act like a bomb!
8. Never pour water into concentrated acid. Pour acid slowly into water, stirring constantly. Mixing acids with water is often exothermic.
9. Concentrated acids and bases are stored in the fume hood.
10. Make sure test tubes containing reactions are pointed away from people, especially when. They are being heated.

PHYSICAL HAZARDS

1. Don't make noise in the laboratory
2. Avoid excessive movement of your body
3. Don't touch any equipment or machinery until it’s needed.

BIOLOGICAL HAZARD:

1. While working with any microbe, avoid contact with skin; use gloves.
2. Be careful when handling experimental animals.
3.Avoid direct contact with any plant, animal or microbe.
4. Whenever working on blood samples, always use disposable containers and avoid contamination.

Chapter 2: Lab Equipment

Introduction to Lab Equipment

Standard Operating Procedures are essential for ensuring the safe and effective operation of laboratory equipment. They provide detailed step-by-step instructions for performing specific tasks, promoting consistency, accuracy, and safety in lab operations.

General Principles for Developing SOPs:

1. Clarity and Detail: Each SOP should be clear and detailed, outlining each step of the procedure to prevent ambiguity.
2. Safety Considerations: Include information on potential hazards, necessary personal protective equipment (PPE), and emergency procedures.
3. Equipment-Specific Instructions: Provide manufacturer guidelines and any lab-specific modifications or protocols.
4. Training and Competency: Ensure that all personnel are trained in the SOPs and demonstrate competency before operating equipment.
5. Review and Update: Regularly review and update SOPs to reflect current practices and technological advancements.

SOPs for Specific Laboratory Equipment:

1. Thermostatic Bath/Water Bath:



A **water bath** is a piece of laboratory equipment made from a container filled with water. It is used to incubate samples in water at a constant temperature over an extended period. Most water baths come with either a digital or an analogue interface, which allows users to set the desired temperature. Some

water baths control temperature by passing a current through a reader. Water baths have various applications including warming reagents, melting substrates, or incubating cell cultures. They are also essential for enabling certain chemical reactions to occur at higher temperatures. A water bath is often the preferred heat source for heating flammable chemicals, as it is safer than using an open flame and prevents the risk of ignition. Water baths are typically used up to **99.9°C**. When temperatures exceed 100°C, alternative heating methods such as **oil baths**, **silicone baths**, or **sand baths** may be used.

Applications/Uses:

1. **Improving solubility**: It is used to improve the solubility of poorly soluble substances.
2. **Melting substances**: Water baths are used for melting certain substances, such as waxes or fats.
3. **Warming reagents**: They help in warming chemical reagents to the desired temperature for reactions.
4. **Facilitating chemical reactions**: Certain chemical reactions that require heat can be enhanced in a water bath.
5. **Incubating cell cultures**: Water baths provide a stable temperature environment for growing and maintaining cell cultures.
6. **Safe heat source for flammable chemicals**: Water baths are used to heat flammable chemicals safely, as they eliminate the risks associated with open flames.

2. Shaker/Mixer:



A **shaker** is a piece of laboratory equipment used to mix, blend, or agitate substances in a tube or flask by shaking them. It is mainly used in the fields of chemistry and biology. The shaker typically contains an oscillating platform that holds flasks, beakers, or test tubes. Though the magnetic stirrer has become

more popular in recent years, the shaker remains the preferred equipment when dealing with larger volumes of substances or when simultaneous agitation is required.

Shakers come in various types, such as orbital shakers (which provide a circular motion) and reciprocating shakers (which provide a back-and-forth motion). These devices are essential for ensuring thorough mixing and consistent results, especially in experiments that require agitation over extended periods.

Applications/Uses:

1. **Homogeneous mixing**: Used for mixing different liquids as well as solid and liquid components, such as nutrient solutions.
2. **Shaking diagnostic kits**: Essential in shaking kits for diagnostic tracing reactions in clinical labs.
3. **Solvent extraction**: Helps in solvent extraction of different phases in separating funnels during chemical investigations.
4. **Agitation of liquid or gaseous phases**: Provides evenly changing agitation for liquid phases (such as nutrient solutions) or gaseous phases in experimental setups.
5. **Coloring and discoloring gels**: Used in processes like gel staining or decolorization in biochemistry and molecular biology.

3. Spectrophotometer:



A **spectrophotometer** is an analytical instrument used to quantitatively measure the transmission or reflection of visible light, UV light, or infrared light. Spectrophotometers measure light intensity as a function of the wavelength of the light source. There are two main classes of spectrophotometers:

1. **Single-beam spectrophotometers**: Measure absolute light intensity.
2. **Dual-beam spectrophotometers**: Measure the ratio of light intensities along two separate light paths, the reference standard and the sample.

Types of spectrophotometers include **microplate spectrophotometers**, **colorimeters**, and **fluorescence microplate readers**. These instruments are widely used in various fields such as **physics**, **molecular biology**, **chemistry**, and **biochemistry**.

Spectrophotometers are commonly used to measure the concentration of substances like proteins, DNA, RNA, growth of bacterial cells, and enzymatic reactions. Some key features to consider when selecting a spectrophotometer include its **dynamic range**, **sample volume capacity**, and the **lamp type and its lifetime**.

Spectrophotometric data can also be used with the **Beer-Lambert Equation** to establish relationships between **transmittance** and **concentration**, and **absorbance** and **concentration**. As spectrophotometers measure a compound's wavelength through its color, a dye-binding substance can be added to induce a color change, which can then be measured. Spectrophotometers are specialized for either UV or visible light wavelength absorbance measurements.

It is a highly accurate, sensitive, and precise instrument, especially for determining color changes in samples.

Applications/Uses:

1. **Biochemical experiments**: Spectrophotometry is crucial for a wide range of biochemical experiments, especially those that involve measuring the concentration of substances.
2. **Measurement of colored compounds**: Used to measure colored compounds in the visible region of light (between 350 nm and 800 nm).
3. **DNA, RNA, and protein analysis**: Commonly used in DNA, RNA, and protein isolation, as well as enzyme kinetics and other biochemical analyses.
4. **Inexpensive and simple method**: This method is convenient and cost-effective for laboratory experiments, offering ease of use with relatively low complexity.

4. Electronic Balance/Analytical Balance:



An **analytical balance** (often called a "lab balance") is a highly precise instrument designed to measure small masses in the sub-milligram range, typically with a sensitivity of 0.1 mg or better. The measuring pan of an analytical balance is enclosed in a transparent case, often referred to as a **draft shield**, to prevent dust from collecting and to eliminate any interference from air currents in the room. This shield ensures accurate measurement by protecting the sample from external environmental factors.

Some advanced analytical balances come with a **mechanically vented safety enclosure**, equipped with uniquely designed **acrylic airfoils** that provide smooth, turbulence-free airflow. This ensures no fluctuations in the balance's readings, enabling the measurement of mass as precisely as 1µg. These balances are often used in sensitive applications where even the smallest change in mass is important.

A **single-pan mechanical substitution balance** maintains a consistent response throughout the balance's capacity by subtracting mass on the same side of the balance beam where the sample is added. This helps to prevent errors from air currents or temperature changes that might otherwise affect measurements.

Applications/Uses:

1. **Academic Laboratories**: Used for precise measurements in educational settings, particularly in chemistry, biology, and physics experiments.
2. **Pharmaceutical Labs**: Critical for measuring small quantities of substances in drug discovery and formulation.
3. **Genomics and Proteomics Labs**: Utilized in molecular biology research for measuring DNA, RNA, or protein samples accurately.
4. **Precision Mass Measurement**: Ideal for applications requiring high resolution, such as in pharmacology, chemistry, and other fields involving small-scale measurements.

5. **Choosing the Right Balance**: When selecting an analytical balance for a lab, factors such as ease of use, resolution, capacity, and specific applications should be considered to ensure the instrument meets the needs of the laboratory.

5. Oven:



laboratory oven is designed to provide uniform temperature throughout the chamber and is used for a variety of applications that require precise heating. Laboratory ovens are essential for processes such as **annealing**, **die-bond curing**, **drying**, **polyimide baking**, **sterilizing**, and other industrial laboratory functions. Typical sizes of laboratory ovens range from **one cubic foot** to **0.9 cubic meters** (32 cubic feet), with temperatures capable of exceeding **340°C**.

Laboratory ovens can be configured in various ways to meet specific needs, including options such as:

- **Clean room environments**,
- **Forced convection**,
- **Horizontal airflow**,
- **Inert atmosphere setups**,
- **Natural convection**, and
- **Pass-through configurations**.

Industries such as **healthcare**, **technology**, **transportation**, and **civil engineering** commonly use laboratory ovens. In civil engineering, for instance, ovens are used for drying **asphalt**, **soil**, **rock**, or **concrete** samples.

Applications/Uses:

1. **Annealing**: Used in the heating and controlled cooling of materials like glass or steel to reduce hardness and increase ductility. High-temperature ovens are essential for this process.
2. **Die-bond curing**: Through drying and baking, laboratory ovens cure substances to harden their chemical composition, commonly used in **polymer research**, **nanotechnology**, and the **semiconductor industry** for creating materials such as epoxies, glues, plastics, and rubbers.
3. **Drying**: Used in environmental, biological, and clinical labs to remove moisture from samples. Gravity convection, forced air, and vacuum ovens are used for this purpose, depending on the specific requirements.
4. **Polyimide baking**: Polyimide is applied in liquid form and then thermally baked in an oven to create thin films or layers for a variety of uses, including **stress buffer coatings**, **redistribution layers**, **adhesion**, and **chip bonding**.
5. **Sterilizing**: Laboratory ovens are used to sterilize equipment and glassware, typically requiring temperatures of at least **160°C**. The sterilization process is carried out in a **hot air oven**, with content monitored at this temperature for **45 to 60 minutes**.

6. Glucometer:



Blood Glucose Meter and Test Strips

A **glucose meter** is a medical device used to determine the approximate concentration of glucose in the blood. It is commonly used for **home blood glucose monitoring (HBGM)** by individuals with **diabetes mellitus** or **hypoglycemia**. The device works by placing a small drop of blood, obtained by pricking the skin with a **lancet**, onto a disposable test strip. The meter reads the strip and calculates the blood glucose level, which is then displayed in units of **mg/dl** or **mmol/l**.

The glucometer is an essential tool for managing blood sugar levels and can be used to track glucose levels throughout the day. Some models allow for the use of **glucose paper strips** that can be dipped into a sample to measure glucose concentration. The device gives quick, accurate results to help individuals monitor their condition regularly.

Applications/Uses:

1. **Monitoring Blood Glucose**: Glucometers are primarily used for measuring blood glucose levels in individuals with diabetes or hypoglycemia.
2. **Fasting Blood Sugar (FBS) Test**: The glucometer is used to measure blood sugar levels during fasting periods, providing key insights into glucose regulation.
3. **Glycosylated Hemoglobin (A1C) Test**: The glucometer can aid in the determination of A1C levels, which reflect blood sugar levels over the past **90 days**.
4. **Managing Type 1 Diabetes**: For individuals with **Type 1 diabetes**, regular blood sugar testing is essential, often recommended **4 to 10 times a day**, including before meals, snacks, exercise, and sometimes during the night.
5. **Managing Type 2 Diabetes**: Individuals with **Type 2 diabetes** may not need to test blood sugar levels daily, especially if they are managing their condition with **non-insulin medications**, **diet**, and **exercise**.

7. PH Meter:



A **pH meter** is a scientific instrument used to measure the hydrogen-ion activity in water-based solutions, indicating their **acidity** or **alkalinity**. The pH meter works by measuring the difference in electrical potential between a **pH electrode** and a **reference electrode**. This difference in electrical potential is directly related to the acidity or pH of the solution. The result is displayed as the corresponding pH value after the voltage between the two electrodes is measured.

The typical pH meter comprises a simple **electronic amplifier**, a pair of electrodes (or alternatively, a **combination electrode**), and some form of **display** calibrated in **pH units**. The most common electrodes used are a **glass electrode** and a **reference**

electrode, although combination electrodes that integrate both functions are also available. The electrodes (or probes) are inserted into the solution being tested.

Calibration of the pH meter is performed using at least two standard **buffer solutions** that span the pH range being measured. Typically, buffers at **pH 4.00** and **pH 10.00** are used to calibrate the device.

Applications/Uses:

1. **Measuring pH of liquids**: Commonly used to measure the pH of liquids, such as water, solutions, and chemicals in various industries.
2. **Measuring pH of semi-solids**: Specialized electrodes allow pH measurement of semi-solid substances like foods.
3. **Body fluids**: Used for determining the pH of different body fluids in healthcare and clinical settings.
4. **Soil measurements**: Employed in agriculture for assessing soil pH, which is critical for determining soil health and suitability for different crops.
5. **Water quality**: Important in municipal water supplies, swimming pool monitoring, and environmental remediation for ensuring water quality standards.
6. **Brewing and fermentation**: Used in the brewing of wine and beer to monitor and control the fermentation process.
7. **Manufacturing applications**: In industries that require precise pH control for product quality and consistency.
8. **Clinical applications**: Essential in healthcare for **blood chemistry** and other diagnostic tests that rely on pH balance.

8. Chromatography:



Chromatography is an analytical technique used to separate a given mixture into its constituents. It involves separating the components, or solutes, of a mixture based on the relative amounts of each solute distributed between a moving fluid stream, called the mobile phase, and a contiguous stationary phase. The mobile phase may be either a Liquid or a gas, while the stationary phase is either solid or a liquid. The most widely

used types of chromatography include gas chromatography (GC), liquid chromatography (LC), and thin layer chromatography (TLC).

Chromatography is extensively applied in chemical analysis, biochemistry, and forensic science to quantify and identify parts of a mixture.

Applications/Uses

1. **Toxicology**: detect benzodiazepine and anabolic steroid use, even at pg./mL levels in urine and nail samples, making it effective for diagnosing use disorders and athlete doping.
2. **Environmental Testing:** Used for detecting pollutants in air, water, and soil samples.
3. **Biochemical Assays:** Used in laboratories for separation, identification, and quantitation of proteins, enzymes, and DNA/RNA.
4. **Forensic Analysis:** Applied to identify substances that are found during crimes, i.e., drugs, poisons, and toxins.

9. Electrophoresis:



Electrophoresis is a basic technique in biochemistry and molecular biology for separating and analyzing macromolecules, like proteins, carbohydrates, and nucleic acids, on the basis of charge and size. Electrophoresis is an excellent technique in a wide range of scientific fields, including genomics, genetics, proteomics, and clinical diagnosis

Common types of electrophoresis include **agarose gel electrophoresis**, **polymerase chain reaction (PCR)**, and **two-dimensional electrophoresis**.

Applications/Uses:

1. **DNA/RNA Fragmentation**: Used for **genetic analysis**, allowing researchers to separate and analyze DNA and RNA samples.
2. **Protein Analysis**: Used to study proteins, enzymes, and other biological macromolecules, often in proteomics research.

3. **Clinical Diagnostics**: Employed for the separation and analysis of biomarkers, such as in **blood serum electrophoresis** for disease diagnosis.
4. **Genetic Research**: Essential for studying gene expression and the identification of genetic markers in research and diagnostic labs.
5. **Forensic Identification**: Used in forensic science for DNA fingerprinting and identifying genetic relationships.

10.CHEMISTRY ANALYZER:



Biochemistry Analyzer is a medical laboratory equipment used extensively in clinical laboratories. It is designed to conduct tests to measure chemicals at various stages of the biological processes on a wide range of test samples from blood, plasma, serum, and cerebrospinal fluid to urine. It helps to study, analyze, and evaluate the characteristics of different body samples and tackle a broad spectrum of medical industry challenges.

PRINCIPAL:

A biochemistry analyzer works by the principle of spectrophotometry, a phenomenon that identifies the reaction of a biological sample with certain reagents. When a light at a particular wavelength is passed through a solution (incident light), some amount of it is absorbed and, therefore, the light that comes out (transmitted light) is diminished. Beer-Lambert law governs the nature of light absorption in a solution. Beer's law states that the amount of transmitted light decreases exponentially with an increase in the concentration of absorbing material (i.e. the amount of light absorbed depends on the concentration of the absorbing molecules). In addition, according to Lambert's law, the transmitted light decreases exponentially with increase in the thickness of the absorbing molecules (i.e. the amount of light absorbed is dependent on the thickness of the medium).

Applications/Uses:

1. It has made it possible to detect and monitor the diseases by analysis of chemical reactions and biological processes taking place inside the patient’s body within less time.
2. It speeds up the analysis of complex biochemical reactions.

3. It helps with performing different kinds of tests at affordable rates.
4. It increases the efficiency of test centers, medical facilities, and diagnostic centers (small, medium, or large).

Chapter 3: Sample drawing Techniques

1. Venipuncture

Definition: Venipuncture refers to the procedure of obtaining intravenous access for intravenous therapy or blood withdrawal. Venipuncture is simply achieved with a butterfly needle, vacuum tube, or syringe. The blood consists of plasma, red blood corpuscles, white blood corpuscles, and platelets. The function of transporting oxygen, nutrients, and body waste is performed by the blood. The blood samples are used in every clinical laboratory on a regular basis for CBC (Complete Blood Count), tests of liver and kidneys, and electrolyte analysis.

Procedure:

1.Patient Preparation: Verify patient identification, educate the patient, and get consent.

2.Site Selection: The sites are usually chosen in the median cubital vein, cephalic vein, or basilic vein.

3.Aseptic Technique: Clean with alcohol swabs and allow it to dry.

4. Need Insertion: Insert the needle 15-30°, aspirate for blood, and take out the needle when obtaining the sample.

5. Post-procedure Care: Squeeze gauze over the site, label the specimen, and dispose of sharps.

Complications:

• Hematoma formation

• Phlebitis

• Nerve damage

• Infection

2. Arterial Blood Gas:

Analysis is an important test for the evaluation of a patient's oxygenation, ventilation, and acid base balance. The radial artery is frequently selected for sampling because of its accessibility and collateral circulation.

Procedure for Radial Artery Blood Sampling:

1.Patient Preparation:

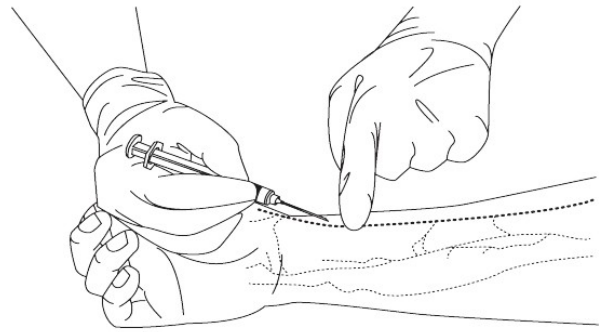
- **Identification:** Verify the patient's identity and describe the procedure to obtain informed consent.
- **Positioning:** Place the patient in a supine position with the wrist extended and supported by a rolled towel.

2.Site Preparation:

- Perform hand hygiene and don non-sterile gloves.
- Clean the puncture site with 70% isopropyl alcohol and let air dry.

3.Sample Collection:

- **Equipment:** Use a pre-heparinized syringe with 20–25-gauge needle.
- **Technique:**
    1. Locate the pulse of the radial artery by palpation
    2. Insert the needle at the 45-degree angle, bevel up, and distal to the palpating fingers.
    3. Allow the syringe to fill spontaneously; do not pull back on the plunger.
    4. Once the desired volume (usually 1–2 mL) is achieved, withdraw the needle immediately



Locate artery and take a sample

4.Post-Procedure Care:

- Use firm pressure to the puncture site immediately with sterile gauze for at least 5 minutes to prevent hematoma formation.
- When bleeding is under control, bandage the site with a sterile dressing.

5.Sample Handling:

- Expel air bubbles from the syringe to prevent alteration of gas values.
- Cap the syringe and agitate the sample gently to determine that it is properly anticoagulated.
- Label the sample appropriately with the patient identification, and transport the sample to the laboratory immediately, preferably being analyzed within 10-15 minutes if stored at room temperature.

Complications

1. Hematoma or excessive bleeding
2. Nerve damage
3. Fainting or a vasovagal response can be prevented by ensuring that the patient is supine (lying down on their back) with feet elevated before beginning the blood drawing

3. Capillary Sampling

Introduction:

Capillary blood sampling is a minimally invasive technique that is utilized when only a small quantity of blood is required, e.g., for glucose testing, hemoglobin measurement, or newborn screening. It is extremely helpful for children and elderly patients where venous access is

difficult. Capillary sampling involves the removal of blood from small vessels close to the skin surface, most commonly from the fingertip, heel, or earlobe.

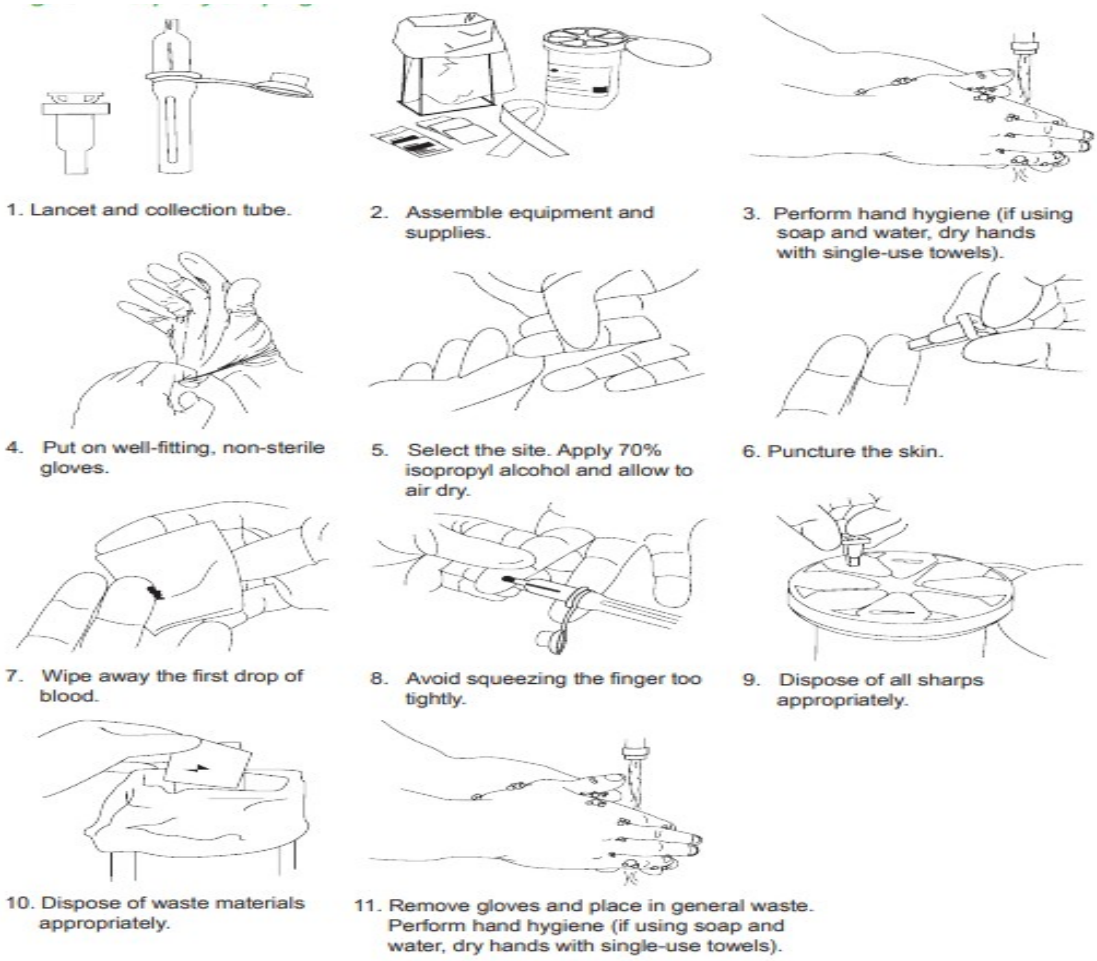
Procedure for Capillary Blood Sampling

1. Patient Preparation:

- Identify and explain the procedure to the patient to obtain their cooperation.
- Select a suitable puncture site:
    1. Adults & older children: Fingertip (middle or ring finger)
    2. Neonates & infants: Heel (medial or lateral plantar surface)
    3. Avoid sites which have been punctured before, swollen, infected, or bruised.

2. Site Preparation:

- Wear gloves and wash hands.
- Prepare the chosen site with 70% isopropyl alcohol and let it air dry.
- Warm the site (if necessary) with a warm compress to increase blood flow.



3. Sample Collection:

- Use a sterile lancet or an automated skin puncture device.
- For fingertip sampling, puncture transverse into the fingerprint lines to allow proper flow.
- For neonatal heel sampling, puncture the appropriate heel area to avoid injury.
- Blot away the first drop of blood to remove tissue fluid contamination.
- Permit blood to enter the collection tube naturally without too much squeezing to avoid dilution with interstitial fluid.

4. Post-Procedure Care:

- Use sterile gauze to control bleeding.
- Dispose of the lancet in a sharp container.
- Label the sample appropriately and transport it immediately.

Capillary sampling is a valuable technique for quick and easy blood sampling, minimizing patient discomfort while providing accurate biochemical analysis.

4.CSF Collection

Cerebrospinal fluid (CSF) tapping, or lumbar puncture, is a medical procedure used to obtain a sample of the fluid surrounding the brain and spinal cord. The fluid serves as a cushion to protect these structures from injury, and lab testing can aid in the diagnosis of certain neurological illnesses.

Procedure for CSF Collection:

1.Preparation:

- **Patient Position:** The patient is asked to lie on one side with the knees folded into the chest and chin down toward the chest. Alternatively, the procedure can be administered in a sitting position leaning forward.
- **Site Preparation**: The region where the lower spine is situated is disinfected with antiseptics for a sterile environment.

2.Anesthesia:

- A local anesthetic medication is applied at the lower spinal area to insensitize the site, thus alleviating pain experienced during treatment.

3.Needle Insertion

- A spinal needle is inserted gently between the lumbar vertebrae into the subarachnoid space containing the CSF.
- In certain situations, imaging guidance, such as fluoroscopy, may be employed to verify correct placement of the needle.

4.CSF Collection:

- The needle is left in place, and a small amount of CSF (usually 1 to 10 milliliters) is aspired into sterile containers for examination.

5.Post-Procedure:

- The needle is withdrawn, and the puncture site is dressed.
- Patients are typically instructed to lie flat for a brief time to minimize the risk of headaches and are advised to take more fluids after the procedure.

Analysis of CSF can be of immense assistance in the diagnosis of such conditions as infections (e.g., meningitis), hemorrhage around the brain (subarachnoid hemorrhage), and other neurologic disorders.

Chapter: 4 (Overview of samples and Sample collection techniques)

The handling and collection of samples are very important in an effort to deliver accurate diagnostic reports.

Types of Samples, Methods of Collection Types of vacutainers and their color coding

1. Whole Blood: Whole blood refers to unseparated blood containing all its components red blood cells (RBCs), plasma, white blood cells (WBCs), and platelets. It is drawn directly into tubes containing anticoagulants to avoid clotting. Most often used for hematological testing.

Collection:

- Collected in tubes containing anticoagulants (e.g., EDTA, heparin) to prevent clotting.
- Used for hematological studies and blood gas analysis.

Applications:

- Complete Blood Count (CBC)
- Blood glucose estimation
- Hemoglobin measurement

Types of vacutainers:

Lavender Top (Purple): Contains EDTA (Ethylenediaminetetraacetic acid) as an anticoagulant. EDTA chelates calcium ions, preventing blood clotting. This tube is primarily used for hematological examinations, including complete blood counts (CBC), erythrocyte sedimentation rate (ESR), and peripheral blood smears.

2. Serum: It is the liquid portion of blood obtained after the clotting. It's made after the removal of cells and clotting factors. It is drawn by filling up blood tubes with no anticoagulants and letting it clot. The sample is centrifuged once clotted to remove the clot and leave behind the serum. Serum is used mostly for all biochemical assays

Collection:

- Blood is drawn into plain (red-top) tubes without anticoagulants.
- Allowed to clot and then centrifuged to separate the serum.

Applications:

- Liver function tests
- Renal function tests
- Hormonal assays (thyroid function tests)

Types of vacutainers:

Red Top: Contains no anticoagulant, allowing the blood to clot naturally. After centrifugation, the clear serum is separated from the clot. This tube is used for serological tests and various biochemical assays.

Gold Top (Serum Separator Tube (SST)): Contains a clot activator and a gel for serum separation. This gel forms a barrier between the clot and serum upon centrifugation, facilitating easier serum collection. Used for a wide range of serological and biochemical tests.

3.Plasma: plasma is a fluid portion of blood obtained by centrifuging whole blood collected with an anticoagulant. Pour in tubes containing anticoagulant to avoid clotting. The plasma is centrifuged off from the blood cells. Plasma is utilized for coagulation tests and other biochemistry.

Collection:

- Blood is collected in tubes containing anticoagulants such as heparin or citrate.
- Centrifugation separates plasma from cellular components.

Applications:

- Coagulation studies (e.g. PTT and APTT)
- Electrolyte measurement
- Protein assays

Types of vacutainers:

Green Top: Contains heparin (either lithium or sodium heparin) as an anticoagulant. It enhances the activity of antithrombin III, inhibiting thrombin and preventing clot formation. This tube is used for plasma determinations in clinical chemistry.

Light Blue Top: Contains sodium citrate as an anticoagulant. Citrate binds calcium ions, inhibiting the coagulation cascade. This tube is essential for coagulation studies, such as PT and APTT.

4.Urine: Urine is a biological liquid sample excreted by the kidneys, containing metabolic waste and analytes. Pour in dry, sterile containers. Midstream urine is the preferred specimen to avoid contamination. Urine test is utilized for renal function testing and diagnosis of systemic or metabolic disease.

Collection:

- Random, first-morning, or 24-hour urine samples are collected depending on the test requirements.

Applications:

- Urinalysis (for glucose, proteins, ketones)
- Kidney function tests
- Drug screening

Container:

While vacutainer tubes are primarily designed for blood collection, specific tubes are available for urine samples. These tubes may contain preservatives to stabilize analytes, depending on the required analysis. It's essential to use the appropriate container as per the test requirements.

5.Cerebrospinal Fluid (CSF): CSF is a clear, colorless fluid that circulates in the brain and spinal cord, providing protection and nourishment. Collected by lumbar puncture in aseptic technique. CSF examination is employed for the diagnosis of neurological infection and disease.

Collection:

- Obtained by lumbar puncture (site: L3-L4 or L4-L5 intervertebral space).

Applications:

- Diagnosis of meningitis (bacterial, viral, or fungal infections)
- **N**eurological disorders (e.g., multiple sclerosis)

- Subarachnoid hemorrhage detection

Types of vacutainers:

CSF collection is typically performed via lumbar puncture and collected into sterile tubes without additives. The collected CSF is then distributed into multiple tubes for various analyses, including biochemical, microbiological, and cytological examinations. Proper handling and prompt processing are vital to maintain sample integrity.

Understanding the specific functions and appropriate applications of each vacutainer tube and anticoagulant ensures the accuracy of laboratory results and the safety of both patients and healthcare professionals.

Anticoagulants and Their Uses

Anticoagulant	Mechanism	Common Uses
EDTA (Ethylenediaminetetraacetic acid)	Binds calcium, preventing clotting	CBC, hemoglobin estimation
Heparin	Inhibits thrombin and other clotting factors	Blood gas analysis, clinical chemistry tests
Citrate	Chelates calcium	Coagulation studies
Fluoride	Inhibits glycolysis	Blood glucose estimation

Chapter: 5 (Biochemical Sample Use for Analysis and their interpretation)

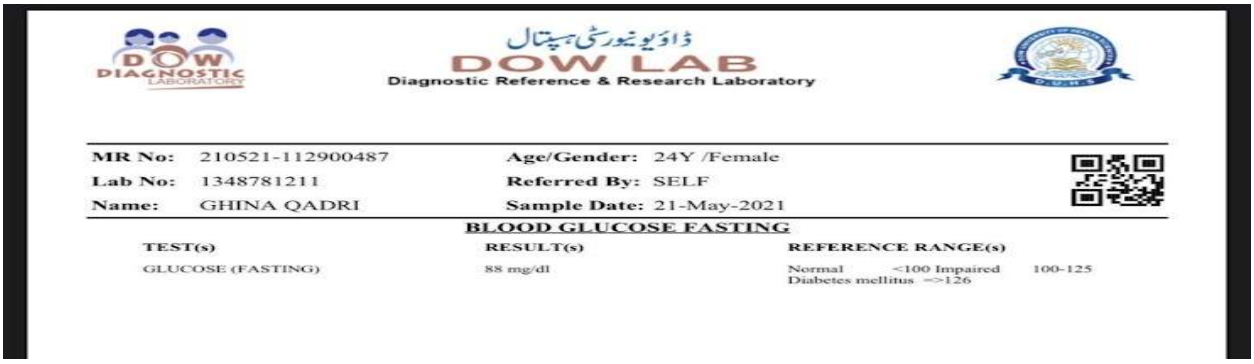
1. Blood Biochemical Analysis

Objective: To analyze blood samples for various biochemical parameters and interpret results.

Sample Use: Blood is a common sample used to assess various biochemical markers like glucose, electrolytes, enzymes, proteins, and lipids.

Parameters and Analysis:

- **Glucose Level**: Blood glucose measurement is used to diagnose diabetes mellitus.
  - **Normal Range**: blood glucose: 70–99 mg/dL (Fasting)
  - **Interpretation**: A higher value indicates hyperglycemia, and lower levels suggest hypoglycemia.



ڈاؤ یونیورسٹی ہسپتال
DOW LAB
Diagnostic Reference & Research Laboratory

MR No:	210521-112900487	**Age/Gender:**	24Y /Female
Lab No:	1348781211	**Referred By:**	SELF
Name:	GHINA QADRI	**Sample Date:**	21-May-2021

BLOOD GLUCOSE FASTING

TEST(s)	RESULT(s)	REFERENCE RANGE(s)
GLUCOSE (FASTING)	88 mg/dl	Normal <100 Impaired 100-125 Diabetes mellitus =>126

- **Hemoglobin (Hb)**: Blood hemoglobin levels can be analyzed for anemia.
  - **Normal Range**: 12–16 g/dL for women, 14–18 g/dL for men.
  - **Interpretation**: Low levels indicate anemia, whereas elevated Hb can indicate dehydration or polycythemia.
- **Liver Enzymes (ALT, AST)**: Measurement of ALT and AST helps in assessing liver function.
  - **Normal Range**: ALT: 7–56 U/L, AST: 10–40 U/L.
  - **Interpretation**: Elevated ALT and AST levels indicate liver damage, such as in hepatitis.

Complete Blood Count (CBC) Report

Example of a Normal Complete Blood Count (CBC) Report

Test	Result	Reference Range
Hemoglobin (Hb)	14.5 g/dL	Male: 13.8 - 17.2 g/dL, Female: 12.1 - 15.1 g/dL
Hematocrit (Hct)	44%	Male: 40 - 52%, Female: 36 - 48%

Red Blood Cell Count (RBCs)	4.8 million/µL	Male: 4.7 - 6.1 million/µL, Female: 4.2 - 5.4 million/µL
Mean Corpuscular Volume (MCV)	88 fL	80 - 100 fL
Mean Corpuscular Hemoglobin (MCH)	30 pg	27 - 33 pg
Mean Corpuscular Hemoglobin Concentration (MCHC)	34 g/dL	32 - 36 g/dL
White Blood Cell Count (WBCs)	7,000 /µL	4,000 - 11,000 /µL
Platelet Count	250,000 /µL	150,000 - 450,000 /µL

Differential Leukocyte Count (DLC) & Interpretation:

Type	Percentage	Normal Range	Interpretation
Neutrophils	60%	40-70%	Normal; suggests no acute bacterial infection
Lymphocytes	30%	20-40%	Normal; adequate immune function
Monocytes	6%	2-8%	Normal; no signs of chronic infections or inflammation
Eosinophils	3%	1-6%	Normal; no indication of allergies or parasitic infections

Basophils	1%	<1%	Normal; no unusual allergic or inflammatory responses

Interpretation of Normal CBC Report

1. Hemoglobin & Hematocrit (Hb & Hct) Normal

- Suggests **no anemia or polycythemia**.
- Normal levels indicate **adequate oxygen-carrying capacity of the blood**.

2. Red Blood Cell Count (RBC) Normal

- Indicates normal **erythropoiesis (red cell production in bone marrow)**.
- No evidence of **anemia (low RBC) or polycythemia (high RBC)**.

3. Red Cell Indices (MCV, MCH, MCHC) Normal

- **MCV (88 fL)**: Normal-sized RBCs (normocytic), ruling out microcytic (iron deficiency) or macrocytic (B12/folate deficiency) anemia.
- **MCH & MCHC**: Indicate normal hemoglobin content within RBCs, ruling out hypochromic or hyperchromic disorders.

4. White Blood Cell Count (WBC) Normal

- Suggests a **healthy immune system with no signs of infection, inflammation, or hematologic malignancy**.
- No **leukocytosis (high WBC, seen in infections) or leukopenia (low WBC, seen in viral infections or bone marrow disorders)**.

5. Platelet Count – Normal

- Ensures **normal blood clotting function**.
- No **thrombocytopenia (low platelets, which can cause bleeding) or thrombocytosis (high platelets, which can lead to clot formation)**

2. Serum Biochemical Analysis

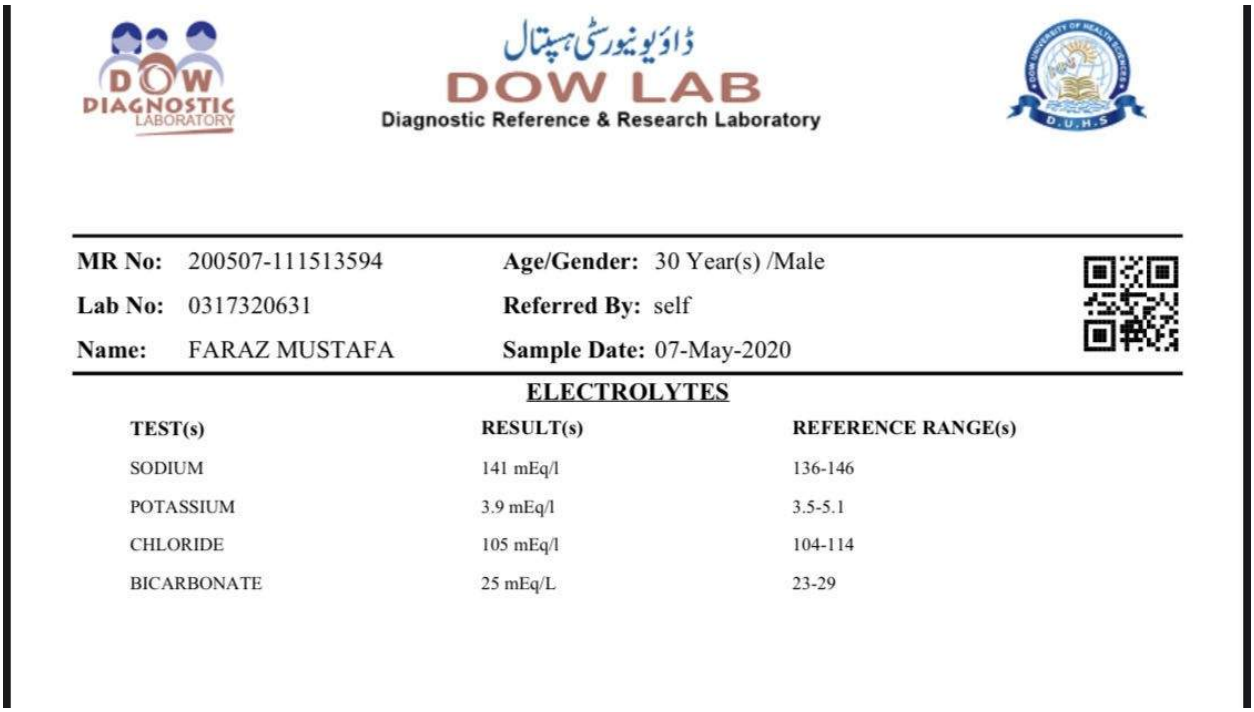
Objective: To perform biochemical tests using serum and interpret the results.

Sample Use: Serum is obtained after blood clotting, and it contains no clotting factors. It's used for most biochemical assays.

Parameters and Analysis:

- **Total Protein**: Serum protein levels can indicate nutritional status and liver function.
  - **Normal Range**: 6–8 g/dL.
  - **Interpretation**: Low levels may suggest malnutrition, liver disease, or kidney disorders, while high levels can indicate dehydration or multiple myeloma.
- **Bilirubin**: Serum bilirubin levels are tested to diagnose liver dysfunction and jaundice.
  - **Normal Range**: Total bilirubin: 0.1–1.2 mg/dL.
  - **Interpretation**: Elevated levels suggest liver disease, hemolysis, or bile duct obstruction.
- **Electrolytes (Sodium, Potassium, Chloride)**: These levels reflect hydration, kidney function, and acid-base balance.
  - **Normal Range**: Sodium: 135–145 mmol/L, Potassium: 3.5–5.0 mmol/L, Chloride: 98–106 mmol/L.
  - **Interpretation**: Abnormal levels indicate conditions such as dehydration, kidney failure, or metabolic disorders.

Example of a Normal Serum Electrolyte Report



ڈاؤ یونیورسٹی ہسپتال

DOW LAB

Diagnostic Reference & Research Laboratory

MR No:	200507-111513594	**Age/Gender:**	30 Year(s) /Male
Lab No:	0317320631	**Referred By:**	self
Name:	FARAZ MUSTAFA	**Sample Date:**	07-May-2020

ELECTROLYTES

TEST(s)	RESULT(s)	REFERENCE RANGE(s)
SODIUM	141 mEq/l	136-146
POTASSIUM	3.9 mEq/l	3.5-5.1
CHLORIDE	105 mEq/l	104-114
BICARBONATE	25 mEq/L	23-29

Example of a Normal Serum Electrolyte Report

Interpretation of Normal Electrolyte Report

1. Normal Sodium (141 mEq/L)

- Maintains **fluid balance, nerve function, and muscle contractions**.
- Indicates **adequate hydration and normal renal function**.

2. Normal Potassium (3.9 mEq/L)

- Essential for **muscle contraction, cardiac function, and nerve signaling**.
- Normal potassium levels suggest **no ongoing renal, GI, or metabolic disturbances**.

3. Normal Chloride (105 mEq/L)

- Maintains **acid-base balance and osmotic pressure**.
- A normal level suggests **no significant metabolic acidosis or alkalosis**.

4. Normal Bicarbonate (25 mEq/L)

- Indicates **proper acid-base homeostasis**.

Normal values suggest **no metabolic acidosis (low HCO_3^-) or metabolic alkalosis (high HCO_3^-)**.

3. Plasma Biochemical Analysis

Objective: To assess plasma samples, which include proteins and clotting factors, to evaluate metabolic and cardiovascular function.

Sample Use: Plasma contains clotting factors, and it's essential for tests involving coagulation.

Parameters and Analysis:

- **Coagulation Studies (Prothrombin Time, Activated Partial Thromboplastin Time)**:
    - **Normal Range**: PT: 11-13.5 seconds, APTT: 30–40 seconds.
    - **Interpretation**: Prolonged PT or APTT indicates clotting disorders, liver disease, or anticoagulant therapy.
- **C-Reactive Protein (CRP)**: Plasma CRP is a marker of inflammation.
    - **Normal Range**: < 3 mg/L.
    - **Interpretation**: Elevated CRP levels indicate acute or chronic inflammation, infection, or autoimmune disorders.

4. Arterial Blood Gases (ABGs)

Objective: To analyze ABGs and interpret respiratory and metabolic conditions.

Sample Use: ABG sampling is used to measure oxygen, carbon dioxide, pH, and bicarbonate in the blood to assess respiratory function and acid-base balance.

Parameters and Analysis:

- **pH**: Reflects blood acidity or alkalinity.
    - **Normal Range**: 7.35–7.45.
    - **Interpretation**: Values below 7.35 indicate acidosis, and values above 7.45 indicate alkalosis.
- **PaO2 (Partial Pressure of Oxygen)**: Indicates oxygenation status.
    - **Normal Range**: 75–100 mmHg.
    - **Interpretation**: Low PaO2 suggests hypoxemia, which could be due to pulmonary disorders or insufficient ventilation.
- **PaCO2 (Partial Pressure of Carbon Dioxide)**: Reflects the effectiveness of ventilation.
    - **Normal Range**: 35–45 mmHg.
    - **Interpretation**: Elevated PaCO2 indicates respiratory acidosis (hypoventilation), while low levels suggest respiratory alkalosis (hyperventilation).
- **HCO3 (Bicarbonate)**: Indicates metabolic component of acid-base balance.
    - **Normal Range**: 22–28 mEq/L.
    - **Interpretation**: Elevated bicarbonate suggests metabolic alkalosis, and low levels indicate metabolic acidosis.

5. Urine Biochemical Analysis

Objective: To assess urine samples for metabolic and renal function.

Sample Use: Urine is often analyzed for the presence of substances that reflect kidney function, hydration status, and systemic conditions.

Parameters and Analysis:

- **Urine Glucose**: Detects glucosuria, which may be indicative of uncontrolled diabetes.
    - **Normal Range**: Negative.
    - **Interpretation**: Presence of glucose suggests hyperglycemia or diabetes mellitus.
- **Proteinuria**: High protein levels in urine can indicate kidney damage.
    - **Normal Range**: Negative to trace.
    - **Interpretation**: Persistent proteinuria may indicate glomerular disease or nephropathy.
- **Urine pH**: Reflects the body's acid-base balance.
    - **Normal Range**: 4.5–8.0.
    - **Interpretation**: An acidic urine pH suggests metabolic acidosis, while an alkaline urine pH may indicate urinary tract infections or renal tubular acidosis.
- **Microscopic Examination**: Examining urine under a microscope can reveal the presence of cells, casts, and crystals.
    - **Interpretation**: Red blood cells may suggest hematuria, while white blood cells indicate infection or inflammation.

Chapter:6

Objectives

Preparation of Solutions - Normal Solution and Normal Saline

At the end of this practical, students should be able to:

1. Define a solution.
2. Differentiate between various types of solutions.
3. Explain the properties of solutes, solvents, and solutions.
4. Demonstrate the calculation and preparation of solutions of different concentrations.

Introduction

Solution:
A homogeneous mixture or single-phase mixture of molecules, atoms, or ions of two or more components is called a solution.

- **Solute:** The substance that dissolves in a solvent is called a solute.
- **Solvent:** A substance that has the ability to disperse a solute is called a solvent.

Types of Solutions

There are three main types of solutions based on the size of solute particles:

1. **True Solution:**
   A solution in which the particle size of the solute is less than 1 nm (nanometer).
   Example: Salt solution.
2. **Colloidal Solution:**
   A solution in which the particle size of the solute ranges from 1 to 200 nm. The particles are dispersed in the solution.
   Example: Protein and polysaccharide solutions.
3. **Crystalloid Solutions:**
   A solution of low molecular weight ions that can pass through a semi-permeable membrane.
   Example: Sodium chloride solution.

Difference Between Emulsion/Suspension and Their Pharmaceutical Applications

Emulsion:
An emulsion is a solution of two immiscible liquids.
Example: Oil in water emulsion.

Suspension:
A suspension is a solution in which the particle size of the solute is greater than 200 micrometers and the particles are suspended in the solution, visible with the naked eye.
Example: Mucaine suspension.

Pharmaceutical Applications:

- **Emulsions:**
  1. Taste and smell of the oil are masked.
  2. The oil becomes more easily digested.
  3. There is uniform distribution of the drug.
- **Suspensions:**
  1. The disagreeable taste of the drug can be masked (e.g., Chloramphenicol palmitate).
  2. Suspensions are generally more stable than solutions.
  3. An ideal dosage form for patients who have difficulty swallowing tablets or capsules (e.g., infants, children, elderly patients).
  4. Drugs are absorbed more rapidly from a suspension than from tablets or capsules.

Preparation for Different Solutions

1. **W/V% Solution:**
   A weight per unit volume solution (W/V%) means the number of grams of solute per 100 mL of solution.
   Example: A 5% W/V solution means 5 grams of solute in 100 mL of solution.
2. **W/W% Solution:**
   A weight per unit weight solution (W/W%) means the number of grams of solute per 100 grams of solution.
   Example: A 5% W/W solution means 5 grams of solute in 95 grams of solvent.
3. **V/V% Solution:**
   A volume per unit volume solution (V/V%) means the number of milliliters of solute per 100 mL of solution.
   Example: A 5% V/V solution means 5 mL of solute in 95 mL of solvent.

Atomic Weight, Molecular Weight, Equivalent Weight, Mole, and Osmole

1. **Atomic Weight:**
   The sum of the number of protons and neutrons in the nucleus of an atom.
2. **Molecular Weight:**
   The sum of atomic weights of atoms presents in a molecule.
3. **Equivalent Weight:**
   The gram molecular weight divided by the acidity, basicity, or valency of the substance.
4. **Mole:**
   The gram molecular weight of a substance.
5. **Osmole:**
   Any substance, molecule, or ion that can exert osmotic pressure.

Molarity, Molality, Normality, Osmolality, and Osmolarity

1. **Molarity (M):**
   The number of moles of a solute in 1 liter (1000 mL) of solution.
2. **Normality (N):**
   The gram equivalent weight of a solute in 1 liter (1000 mL) of solution.
3. **Osmolality:**
   The number of particles of solute per unit weight of water, irrespective of the size and nature of the particles. Units are mmol/Kg of water.
4. **Osmolarity:**
   The number of particles of solute per liter of solution. Units are mmol/L.

Practical 1: Preparation of a 5% W/V Solution of Glucose

Objective:
To prepare a 5% W/V solution of glucose in distilled water.

Principle:
A W/V% solution means the weight (in grams) of a solute in 100 mL of solution. Glucose is a hydrophilic (polar) molecule that easily dissolves in water. A 5% glucose solution is prepared by dissolving 5 grams of glucose in 100 mL of distilled water.

Requirements:

1. Glucose
2. Distilled water
3. Pipette
4. Flask
5. Weighing machine

Procedure:

1. Weigh 5 grams of glucose using the weighing machine.
2. Take a small amount of distilled water using the pipette and add it to a flask.
3. Add the weighed glucose to the flask and mix the two contents until a homogeneous solution is formed.
4. Adjust the volume of the solution to 100 mL using distilled water.

Result:

Interpretation:

Practical 2: Preparation of a 0.9% Solution of Sodium Chloride (NaCl)

Objective:
To prepare a 0.9% NaCl solution (Normal Saline) in distilled water.

Principle:
A W/V% solution means the weight (in grams) of a solute in 100 mL of solution. Sodium chloride (NaCl) is a hydrophilic (polar) molecule that dissolves easily in water. A 0.9% NaCl solution is prepared by dissolving 0.9 grams of sodium chloride in 100 mL of distilled water.

Requirements:

1. Sodium chloride (NaCl)
2. Distilled water
3. Pipette
4. Flask
5. Weighing machine

Procedure:

1. Weigh 0.9 grams of sodium chloride using the weighing machine.
2. Take a small amount of distilled water using the pipette and add it to a flask.
3. Add the weighed sodium chloride to the flask and mix the contents until a homogeneous solution is achieved.
4. Adjust the volume of the solution to 100 mL using distilled water.

Result:

Interpretation:

Chapter 7: Basic Equipment and Techniques

1. Centrifugation:



Introduction

Centrifugation is a basic method in biochemistry that employs centrifugal force to separate suspended particles according to their shape, size, and density. This technique is essential for separating cellular components, refining biomolecules, and studying macromolecular interactions.


Centrifugation of blood

Principle

Centrifugation relies on sedimentation, a process that separates particles by size, shape, and density using centrifugal force.

Types of Centrifuges

1 Low-Speed Centrifuge: A centrifuge operating at 4,000–6,000 RPM (**Revolutions Per Minute**). Designed for routine separations where high forces are unnecessary

- **Speed:** 4,000–6,000 rpm
- **Applications:**
    - Blood sample sedimentation

2 High-Speed Refrigerated Centrifuge: A centrifuge reaching up to 25,000 RPM. Equipped with cooling systems to prevent sample degradation.

- **Speed:** Up to 25,000 rpm
- **Applications:**
    - Microbial cell harvesting
    - Organelle isolation (mitochondria, lysosomes)

3 Ultracentrifuge: A high-performance centrifuge capable of >70,000 RPM. Used for **nanoscale separations** and **analytical studies**.

- **Speed:** >70,000 rpm

- **Applications:** Used for nanoscale separations and analytical studies.

Applications in Biochemistry

- **Subcellular Fractionation:**
    - Isolation of mitochondria, ER, Golgi apparatus
- **Clinical Diagnostics:**
    - Lipoprotein profiling (LDL, HDL separation)
    - Virus purification (e.g., influenza, HIV)
- **Proteomics & Genomics:**
    - Exosome isolation for biomarker studies
    - DNA/RNA purification
- **Pharmaceutical Research:**
    - Antibody purification via affinity centrifugation
    - Vaccine development (virus-like particle isolation)

2. Spectroscopy



Introduction

Spectroscopic techniques employ electromagnetic radiation (light) to interact with matter, probing molecular features to determine consistency or structure. They utilize UV, visible, infrared, or fluorescence light for the analysis of biomolecules.

Principle

Spectroscopy measures absorption or emission of electromagnetic radiation by biomolecules, where each molecule absorbs or emits light at specific wavelengths due to quantum transitions between energy levels.

Application/Use

- Identification and quantification of biomolecules (proteins, nucleic acids).
- Determining molecular structure and interactions.
- Clinical diagnostics (e.g., DNA concentration, protein assays).

3. Electrochemistry

Introduction

Electrochemical techniques measure chemical species through their electrical properties, employing electrodes and their interactions with ions or molecules in solution.

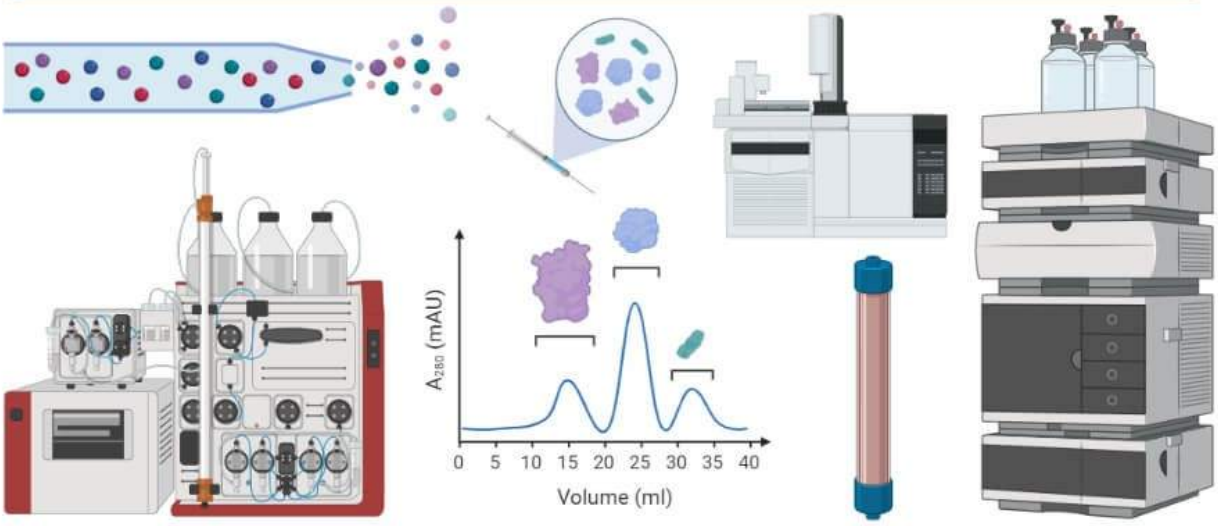
Principle

Ion-selective electrodes (ISEs), like the pH electrode, utilize ion-permeable membranes to measure specific ions. The measured current or voltage is directly proportional to the ion concentration.

Application/Use

- Clinical measurements (glucose monitoring via biosensors).
- Analysis of electrolyte concentrations (Na^+, K^+, Cl^-).
- Determining dissolved oxygen levels in clinical biochemistry.

4. Chromatography



Introduction

Chromatography is a cornerstone technique in biochemistry, enabling the separation, identification, and purification of complex mixtures based on differential interactions between stationary and mobile phases. Its versatility spans clinical diagnostics, pharmaceutical research, and proteomics.

Core Principles

1. **Differential Affinity:**
   Separation occurs due to variations in solubility, charge, size, or adsorption capacity of analytes.
2. **Key Components:**
   - **Stationary Phase:** Solid (e.g., silica gel) or liquid-coated support.
   - **Mobile Phase:** Liquid or gas carrying the sample.

Major Chromatographic Techniques

1. Size Exclusion Chromatography (SEC)

- **Principle:** Separates molecules by size using porous beads. Larger molecules elute first (shorter path).
- **Applications:**
  - Protein purification (e.g., separating hemoglobin from albumin)

- Polymer analysis

2. Ion Exchange Chromatography (IEX)

- **Principle:**
  - Cation exchange: Negatively charged resin binds cations (e.g., Na^+).
  - Anion exchange: Positively charged resin binds anions (e.g., Cl^-).
- **Applications:**
  - DNA/RNA purification
  - Enzyme isolation (e.g., lactate dehydrogenase)

3. Partition Chromatography

- **Principle:** Separation based on solubility differences in polar (aqueous) vs. non-polar (organic) solvents.
  Example: Paper chromatography for amino acids
- **Applications:**
  - Urine screening for inborn metabolic errors (e.g., phenylketonuria)

4. Adsorption Chromatography

- **Principle:** Obsolete but foundational Relies on analyte affinity to adsorbent surfaces (e.g., alumina).
- **Legacy Use:** Early drug purification (e.g., digitalis glycosides)

5. Electrophoresis

Introduction

Electrophoresis is a powerful analytical technique that separates charged molecules (proteins, nucleic acids, and lipoproteins) based on their mobility in an electric field. Widely used in clinical diagnostics, genetic research, and proteomics, it provides critical insights into disease mechanisms and biomolecular interactions.

Core Principles

Mechanism of Separation

- **Charge-Mass Ratio:** Mobility depends on the net charge, size, and shape of the molecule.
- **Electric Field Strength:** Higher voltage accelerates migration but may cause heat-induced denaturation.
- **Support Medium:** Agarose (for large DNA/RNA) or polyacrylamide (for small proteins) gels act as molecular sieves.

Major Electrophoretic Techniques

1 Protein Electrophoresis

- **Clinical Use:**
    - Serum Protein Electrophoresis (SPE): Diagnoses multiple myeloma (M band in γ-region) or nephrotic syndrome (↓ albumin, ↑ α_2-globulin)
    - CSF Analysis: Detects oligoclonal bands in multiple sclerosis

2 Hemoglobin Electrophoresis

- **Diagnosis of Hemoglobinopathies:**
    - HbS (Sickle Cell): Migrates slower than HbA at pH 8.6 due to valine substitution
    - HbA1c: Separated from HbA0 to monitor diabetes

3 DNA/RNA Electrophoresis

- Agarose Gel: Separates nucleic acids by size (ethidium bromide staining)
- **Applications:** PCR product analysis, genotyping

6. Enzymology

Introduction

Enzymology studies enzymes, catalysts that accelerate biochemical reactions. It includes the examination of enzyme kinetics and their catalytic mechanisms.

Principle

Enzymes catalyse biochemical reactions by reducing activation energy. Kinetic studies measure the rate of reactions, determining enzyme activity and reaction mechanisms (Michaelis-Menten kinetics).

Application/Use

- Clinical diagnosis (enzyme levels indicating liver or heart disease).
- Drug development (inhibition kinetics).
- Metabolic studies (enzyme function and regulation).

7. Immunological Tests (Hepatitis, PCR, Pregnancy)

Introduction

Immunological tests detect antigens or antibodies specific to diseases or physiological conditions, involving immune reactions such as antigen-antibody interactions or nucleic acid amplification (PCR).

Principle

- **Hepatitis tests:** Detect viral antigens or antibodies.
- **PCR (Polymerase Chain Reaction):** Amplifies specific DNA sequences for pathogen detection.
- **Pregnancy tests:** Detect human chorionic gonadotropin (hCG) hormone using antibody-based assays.

Application/Use

- Diagnosis of infectious diseases (Hepatitis B/C).
- Detection and quantification of viral load (HIV, COVID-19).
- Confirming pregnancy through rapid, accurate testing.

8. Autoanalyzer Application

Introduction

Autoanalyzer are automated analytical instruments used in clinical biochemistry laboratories to rapidly and accurately perform multiple biochemical assays simultaneously.

Principle

Autoanalyzer’s use automated systems integrating sampling, reagent mixing, incubation, reaction measurement, and data analysis. Biosensors incorporated in analysers detect analytes like glucose via specific enzymatic reactions.

Application/Use

- Clinical chemistry analyses (blood glucose, electrolytes, enzymes).
- Monitoring chronic diseases (diabetes, kidney diseases).
- High-throughput laboratory testing.

Advanced Biochemical Techniques

1.Protein purification

Introduction

Protein purification isolates specific proteins from complex mixtures based on differences in physical and chemical properties like size, charge, solubility, and affinity for ligands.

Principle

Methods like chromatography (ion exchange, affinity, gel filtration), precipitation, and centrifugation exploit protein properties for separation. Each method selectively enriches desired proteins.

Application/Use

- Preparation of proteins for clinical assays.
- Research into protein structure/function relationships.
- Pharmaceutical applications (purification of therapeutic proteins).

2. Homogenization

Definition:
Homogenization is a mechanical or chemical process used to break open cells to release their internal components.

Principle:
Homogenization involves breaking a sample into uniform, smaller components to create a **homogeneous mixture**, while preserving its molecular composition. This is achieved through **mechanical forces** such as **shearing**, **cavitation**, and **turbulence**:

- **Shearing**: Fluid layers move at different speeds, exerting friction that breaks particles.
- **Cavitation**: Rapid pressure changes create vapor cavities; their collapse generates shockwaves that fragment particles.

- **Turbulence**: High-speed flow creates eddies that further reduce size and disperse sample contents

Typical devices include bead mills, rotor-stator mixers, ultrasonic (sonicator) homogenizers, and high-pressure systems.

Uses:

Cell and tissue lysis: Preparing samples for nucleic acid extraction or protein analysis.

Food processing: Homogenizing milk or sauces to create stable emulsions.

Biotech & molecular biology: Efficiently disrupting cells (e.g., yeast, bacteria) for downstream studies.

3. LDH (Lactate Dehydrogenase) Purification

Definition:
LDH purification involves isolating the enzyme lactate dehydrogenase, which catalyzes the interconversion of lactate and pyruvate.

Principle:
LDH can be separated using techniques like ammonium sulfate precipitation, gel filtration, and ion-exchange chromatography, based on its size and charge.

Uses:

- **Enzyme Characterization:**
  - **Enables kinetic analysis and evaluation of structure–function relationships.**
- **Educational Training:**
  - **Provides hands-on experience with key techniques: fractionation, chromatography, spectrophotometry, and electrophoresis.**
- **Comparative Analysis:**
  - **Purified LDH is compared with commercial samples to assess purity and performance characteristics.**

4. Enzyme Assays

Introduction

Enzyme assays measure enzyme activity by detecting the amount of product formed or substrate consumed over time under controlled conditions.

Principle

Assays generally use substrates converting into detectable products through enzyme-catalysed reactions. Often assays use spectrophotometry, exploiting absorbance changes due to the formation of products like NADH.

Application/Use

- Measuring enzyme kinetics.
- Clinical diagnostics (e.g., glucose assays).
- Determining enzyme concentration and activity.

5. Gel Filtration (Size-Exclusion Chromatography)

Introduction

Gel filtration separates molecules based on size using porous particles as a molecular sieve.

Principle

Large molecules excluded from the porous particles pass faster through the column, eluting first. Smaller molecules enter the pores, slowing their migration, and eluting later.

Application/Use

- Protein purification.
- Molecular weight determination.
- Desalting and buffer exchange.
- Separation of biomolecules such as enzymes, nucleic acids, polysaccharides.

6. Protein Crystallography

Introduction

Protein crystallography determines the three-dimensional structure of proteins at atomic resolution through X-ray diffraction of protein crystals.

Principle

Proteins crystallized into ordered lattices diffract X-rays, producing patterns analysed computationally to deduce molecular structures. Crystallization conditions (pH, temperature, ionic strength) critically influence crystal formation.

Application/Use

- Determining protein structures.
- Understanding protein function and interactions.
- Drug design and structural biology research.

7.Topic: Enzyme Linked Immunosorbent Assay (ELISA)

Introduction

ELISA is an immunoassay technique for detecting and quantifying soluble substances such as proteins, antibodies, hormones, and antigens. It employs antibodies labelled with an enzyme that catalyses a colorimetric reaction upon binding to a specific antigen, producing measurable color changes in the presence of substrates.

Principle

ELISA involves immobilizing antigens on a solid phase (microtiter plate), and then a labelled antibody (enzyme-linked) binds specifically to the antigen. Upon addition of substrate, the enzyme catalyses a colour change. The intensity of the colour correlates directly with the concentration of antigen present.

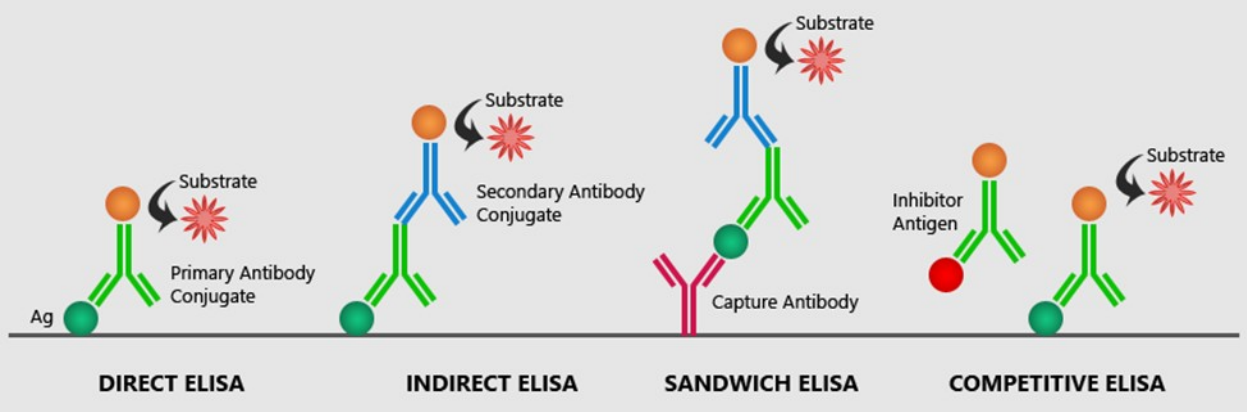
Types of ELISA

1. Triple Antibody Sandwich ELISA (Indirect ELISA)

This method identifies antibodies in patient blood by layering reagents in sequence. Used commonly to detect antibodies against infectious diseases, like hepatitis B virus (HBV), without using live viruses, ensuring safety in diagnostics.

2. Double Antibody Sandwich ELISA (DAS ELISA)

Widely utilized in diagnostics for its simplicity, DAS ELISA immobilizes antigens using a primary antibody on a solid phase, detected with a labelled secondary antibody (enzyme-conjugated), leading to a measurable colour change.



Application/Use

- Diagnosing infectious diseases, such as hepatitis B.
- Ensuring safety in blood transfusions.
- Agricultural diagnostics, detecting viruses in plants (e.g., Potato leafroll virus).
- Clinical diagnosis for hormone levels, drug monitoring, and detecting allergens.

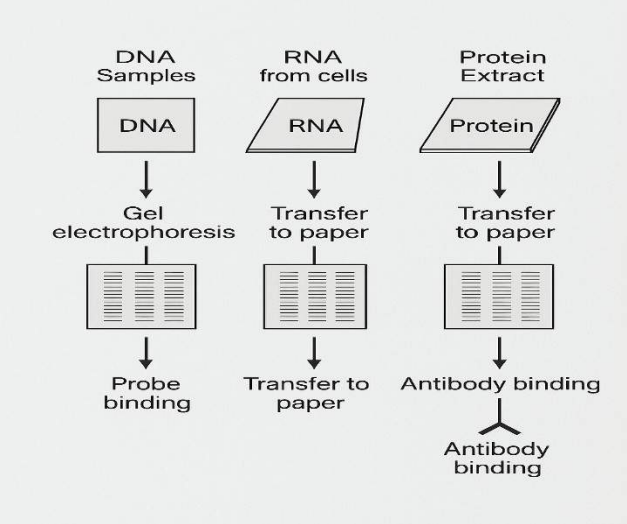
8.Blotting Techniques: Southern, Northern, and Western Blotting

1. Introduction

Blotting techniques are molecular biology tools used to detect specific DNA, RNA, or protein molecules from complex samples. These methods involve

- Electrophoresis to separate molecules by size.
- Transfer of separate molecules to a membrane.
- Detection using specific labeled probes or antibodies.

The three main blotting techniques include:



- **Southern Blot** – for DNA
- **Northern Blot** – for RNA
- **Western Blot** – for Proteins

1. Southern Blotting

Introduction

Southern blotting transfers DNA from gels to membranes for probing specific DNA sequences.

Principle

DNA fragments separated by gel electrophoresis are denatured, transferred onto a membrane (nitrocellulose or nylon), and hybridized with labelled DNA probes complementary to target sequences.

Application/Use

- Gene identification.
- Genetic fingerprinting.
- Detection of related genes across species ("Zoo blotting").

2. Northern Blotting

Northern Blotting

Introduction

Northern blotting identifies specific RNA sequences transferred from gels to membranes.

Principle

RNA separated under denaturing conditions via gel electrophoresis is transferred onto a membrane and hybridized with labelled gene probes specific to target mRNA sequences.

Application/Use

- mRNA detection and quantification.
- Measuring relative mRNA transcript abundance.
- Determining transcript sizes.

3. Western Blotting

Introduction

Western blotting involves the transfer of proteins separated by electrophoresis onto membranes, detected using specific antibodies.

Principle

Proteins electrophoretically transferred to nitrocellulose membranes are probed with specific primary antibodies, followed by enzyme-labelled secondary antibodies, revealing target proteins through a substrate-driven colour reaction.

Application/Use

- Protein identification and quantification.
- Confirming specific protein expression.
- Diagnostic testing for disease markers.

CHAPTER 8: CHARBOHYDRATE

MOLISCH'S TEST

(A General Test For identification of Carbohydrates)

PRINCIPLE:

Molisch's test is a sensitive chemical test, named after Austrian botanist Hans Molisch, based on the dehydration of the carbohydrate by a strong acid like sulfuric acid or hydrochloric acid to produce an aldehyde, which condenses with two molecules of a phenol (usually α-naphthol, though other phenols such as resorcinol and thymol also give colored products), resulting in a purple colored ring at the junction two solution.

REQUIREMENTS:

1. Freshly prepared Alpha Naphthol solution
2. H_2SO_4 (conc.)
3. Clean and dry test tubes
4. Pipettes

PROCEDURE:

1. Take 5 ml of given solution in a test tube and add 2-3 drops of ethanolic Alpha Naphthol (Molisch reagent) and mix.
2. Add 2 ml of H_2SO_4 (conc.) along the walls of the test tube, avoid the mixing of the contents of solution.
3. Allow to stand for 1-2 min.

OBSERVATIONS:
Purple ring appears at the junction of two solutions.

RESULT:

INTERPRETATION:

IODINE TEST

(For identification of Polysaccharides)

PRINCIPLE:
Starch or polysaccharide when reacts with Iodine solution give purple/blue color. The reaction is only physically association where I_2 traps in the coiled structure of polysaccharide. The test is positive only with polysaccharides. Glycogen also gives redish brown color with Iodine, but the intensity of color is lower.

REQUIREMENTS:

1. Test solution
2. Iodine solution
3. Clean and Dry test tubes
4. Pipettes

PROCEDURE:

1. Take 2 ml test solution in a test tube.
2. Add 1-2 drops of iodine solution.
3. Observe the color of the solution.

OBSERVATIONS:

RESULT:

BENEDICT TEST

(For identification of reducing and non-reducing sugar)

PRINCIPLE:
When Benedict's solution and simple carbohydrates are heated, the color of solution changes to orange red/brick red. This reaction is caused by the reducing property of simple carbohydrates. The copper (III) ions in Benedict's solution are reduced to Copper (I) ions, which causes the color change. The color varies from green to dark red (brick) or rusty brown, depending on the amount and type of sugar.

REQUIREMENTS:

1. Given solution of sugar
2. Benedict's reagent
3. Test tubes along with holder
4. Pipettes

PROCEDURE:

Preparation of benedict's reagent:
Anhydrous sodium carbonate= 100gm
Sodium citrate = 173gm
Copper (II) sulphate pentahydrate =17.3 gm

Test PROCEDURE:

1. Take 5 ml of Benedict's reagent in a clean dry test tube.
2. Add 3 ml of given sample.
3. The solution is then heated gradually for 3-5 minutes.
4. Observe for color change in the solution.

OBSERVATIONS:
The color of solutions turns to reddish/orange color indicates the presence of reducing sugar.

RESULT:

INTERPRETATION:

Thymol blue test: is the use of the chemical compound **thymol blue** as a **pH indicator** to determine the acidity or alkalinity of a solution

BARFOED'S TEST

(To distinguish monosaccharide from disaccharide)

PRINCIPLE:

Monosaccharides usually react in about 1-2 minutes while the reducing disaccharides take much longer time between 7-12 minutes to react with the reagent. Brick red color precipitates are obtained due to formation of cuprous oxide, Cu_2O.

REQUIREMENTS:

1. Test solution
2. Barfoed's reagent: cupric acetate in 1% acetic acid
3. Test tubes along with holder

4. Pipettes

PROCEDURE:

1. Take 1 ml of test sample in dry test tube.
2. Add 2 ml of Barfoed's reagent.
3. Heat the solution and wait for the development of brick red precipitate.
4. Note the time taken to develop the color.

Note:
Positive Barfoed's test: development of brick red color ppts within 3-5 minutes
Negative Barfoed's test: absence of red color
Reducing disaccharides also give positive Barfoed test on prolong heating

OBSERVATIONS:

RESULT:

SELIWANOFF TEST

(To distinguish aldoses from ketoses)

PRINCIPLE:
On treatment with conc. Acid, ketoses are dehydrated more rapidly to give furfural derivatives and on condensation with resorcinol give cherry red complex. The test is positive with fructose, sucrose and other ketoses.

REQUIREMENTS:

1. Test solution
2. Seliwanoff’s reagent (0.5% resorcinol in 3N HCl)
3. Water bath
4. Clean & Dry test tubes
5. Pipettes

PROCEDURE:

1. Take 1 ml of test solutions in a test tube.
2. Add 3 ml Seliwanoff’s reagent.
3. Keep the test tube in water bath/heat for 1–2 minutes.
4. Observe the color of the solution.

OBSERVATIONS:

RESULT:

OSAZONE TEST

(Osazone crystals to identify reducing sugars)

Object:
Osazone crystals to identify reducing sugars.

Principle:
When reducing sugar reacts with phenyl hydrazine ($C_6H_5NHNH_2$) at boiling temperature and constant pH they form osazone crystals of reducing sugar. Phenyl hydrazine reacts with carbonyl compound in

neutral or slightly acidic condition to give phenyl hydrazone which is highly soluble. On heating, the hydrazone reacts with further phenyl hydrazine molecules, the condensation products formed are insoluble which precipitate out as crystals.

Reagents:

1. Phenyl hydrazine hydrochloride compound
2. Sodium acetate salt
3. Glacial acetic acid

Procedure	**Observation**	**Inference**
1. Take 5 ml carbohydrate solution in a test tube.	Needle shape crystal – Glucose or fructose is present	
2. Add 10 drops of glacial acetic acid.	Hedgehog or pincushion with pin or flower shaped crystal – Lactose is present	
3. Add a large knife point of phenyl hydrazine.	Sunflower / petal shaped crystals – Maltose is present	
4. Add two large knife points of sodium acetate.		
5. Mix it thoroughly.		
6. Boil the solution for 45 minutes.		
7. With the help of clean glass rod take out some crystals on a glass		

Procedure	Observation	Inference
slide and observe the shape under microscope.		



Needle shaped crystals (Glucose)



Dense ball needle shaped crystals (Arabinose)



Needle shaped crystals (Fructose)



Fine-long needle shaped crystals (Xylose)



Needle shaped crystals (Mannose)



Sun flower shaped crystals (Maltose)



Balls with thorny edge shaped crystals (Galactose)

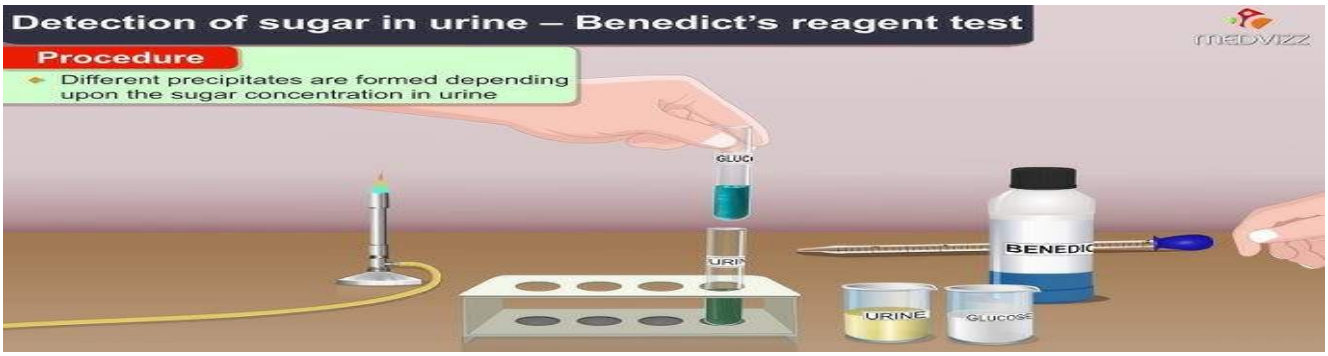


Cotton ball shaped crystals (Lactose)

DIABETES MELLITUS TEST

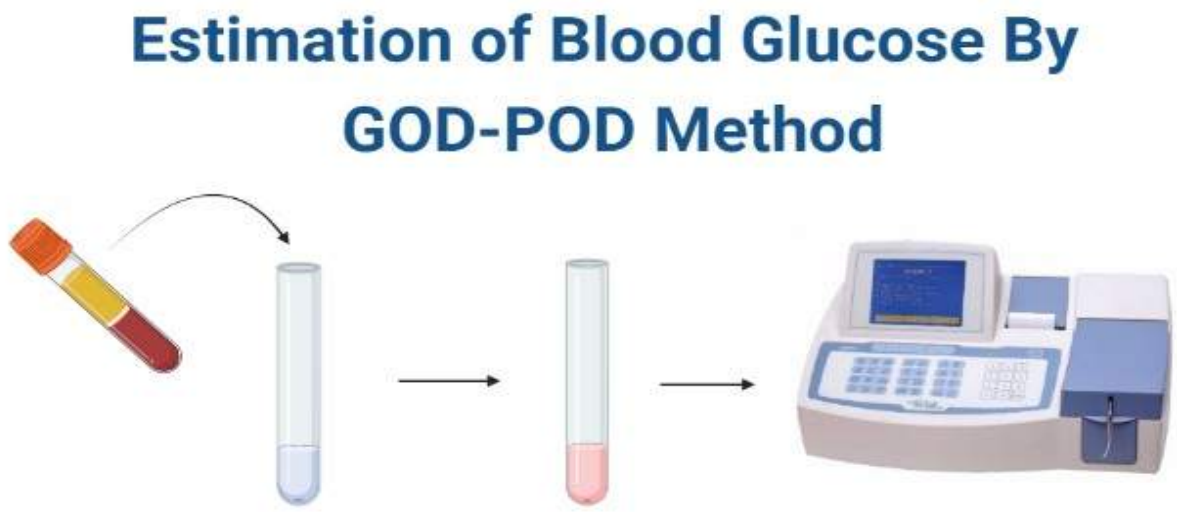
1. Urine Glucose Test

Section	Details
Purpose	To detect the presence of reducing sugars (like glucose) in the urine.
Principle	Benedict's reagent is used. In the presence of reducing sugars, the copper(II) ions are reduced to copper(I) oxide, forming a colored precipitate. The color change (blue to green, yellow, or brick-red) reflects the sugar concentration..
Materials	Benedict's reagent (qualitative), test tubes, water bath or spirit lamp, Pasteur pipette, fresh urine sample.
Method	1. Pipette **5 mL** Benedict's reagent into a clean tube. 2. Add **8–10 drops** of urine; mix gently. 3. Heat to gentle boil for **2–3 min**; cool to room temperature. 4. Compare colour/precipitate against the semi-quantitative scale supplied with the reagent.
Interpretation	Blue: No sugar Green to yellow: Mild presence Orange to red: High sugar level
Clinical Note	Positive result implies plasma glucose above renal threshold (~180 mg/dL); confirm with plasma-based test.



2. Random Blood Sugar (RBS)

Section	Details
Purpose	To estimate glucose at any time of day.
Principle	**Glucose oxidase (GOD)** converts glucose → gluconic acid + H_2O_2. **Peroxidase (POD)** couples H_2O_2 with chromogen (e.g., 4-aminophenazone + phenol) → quinone-imine dye. Absorbance at 505 nm ∝ glucose concentration. Glucose + O_2 —(Glucose Oxidase)→ Gluconic acid + H_2O_2 H_2O_2 + Chromogen —(Peroxidase)→ Colored product + H_2O
Sample	Venous blood in fluoride/oxalate tube (inhibits glycolysis).
Method	1. Centrifuge; separate plasma. 2. Mix **10 µL** plasma with **1 mL** reagent (kit). 3. Incubate at 37 °C for **10 min**. 4. Read absorbance at 505 nm; calculate with calibrator.
Reference Range	< 140 mg/dL (7.8 mmol/L) in a non-fasting state.



3. Fasting Blood Sugar (FBS)

Section	Details
Purpose	Baseline glycaemia after ≥ 8 h fast.
Principle & Reagent	GOD-POD reaction used for RBS; fasting minimises post-prandial variability.
Method	1. Ensure 8–12 h overnight fast. 2. Collect venous blood (fluoride vial); separate plasma. 3. Run on auto-analyser as for RBS.
Interpretation (ADA 2025)	•70–99 mg/dL (fasting)

4. Glycated Haemoglobin (HbA1c)

Section	Details
Purpose	Reflects average plasma glucose over preceding 8–12 weeks.
Principle	Non-enzymatic glycation produces HbA1c. **Cation-exchange High-Performance Liquid Chromatography** separates HbA1c from other fractions; detector quantifies HbA1c as % of total Hb.
Sample	Whole blood in EDTA tube.
Method (IFCC-aligned)	1. Lyse RBCs with hemolysate buffer. 2. Inject sample into HPLC column at 20–25 °C. 3. Elute haemoglobin fractions with stepwise ionic-strength gradient. 4. Integrate HbA1c peak area; express as % or mmol/mol (IFCC units).
Interpretation	• *Non-diabetic*: < 5.7 % (< 39 mmol/mol) • *Pre-diabetes*: 5.7–6.4 % • *Diabetes*: ≥ 6.5 % (repeat to confirm).

5. Oral Glucose Tolerance Test (OGTT)

Section	Description
Purpose	Gold-standard assessment of whole-body glucose handling; diagnosis impaired glucose tolerance, type 2 DM, and gestational DM.
Principle	After fasting, the patient ingests **75 g** anhydrous glucose. Plasma glucose is measured enzymatically (hexokinase or GOD-POD) at baseline and 120 min. Failure to suppress glucose indicates

	impaired insulin action/secretion.
Patient Prep	• Normal diet & activity ≥3 days prior. • Overnight fast **8–14 h**. • No smoking/exercise during the test.
Method (WHO protocol)	1. Draw fasting venous blood into a fluoride tube → plasma (0 min). 2. Give **75 g** glucose in 250–300 mL water (1.75 g kg^{-1} in children, max 75 g). 3. Patient drinks within 5 min; remain seated. 4. Draw a second sample at **120 min**.
Interpretation (venous plasma)	• **Normal** < 140 mg/dL (<7.8 mmol/L) • **Impaired Glucose Tolerance** 140–199 mg/dL (7.8–11.0 mmol/L) • **Diabetes Mellitus** ≥ 200 mg/dL (≥11.1 mmol/L) *Gestational DM*: ≥153 mg/dL (8.5 mmol/L) at 2 h.

Estimation of Total Cholesterol (Enzymatic Colorimetric CHOD-PAP)

Aim

To estimate **total cholesterol** in serum/plasma using the cholesterol oxidase–peroxidase (CHOD-PAP) enzymatic colorimetric method and report in **mg/dL**.

Principle

1. **Hydrolysis**
   Cholesteryl esters + H_2O — (**cholesterol esterase**) → **Cholesterol** + Fatty acids.
2. **Oxidation**
   Cholesterol + O_2 — (**cholesterol oxidase**) → **4-Cholesten-3-one** + **H_2O_2**.
3. **Chromogen formation**
   H_2O_2 + **4-aminoantipyrine** + **phenol** —(**peroxidase**)→ **Quinoneimine dye** (stable colored red complex).

   Note: Cholesterol standard concentration: 200 mg/dl

Specimen

- **Sample:** Serum (preferred) or heparinized plasma.
- **Pre-analytical:** 10–12 h overnight **fast** recommended to minimize post-prandial lipemia.
- **Stability:** Follow your lab/kit instructions (refrigerated, protected from light where applicable).

Procedure (endpoint; manual bench template)

Label three tubes: **Blank (B)**, **Standard (S)**, **Test (T)**. Add as follows, mix, and incubate **5 min at 37 °C**; then read absorbance of **S** and **T** against **B**:

Estimation of Serum Cholesterol – Procedure

Reagents	Blank	Standard	Sample
Working reagent	1000 µl	1000 µl	1000 µl
Standard	–	10 µl	–
Sample	–	–	10 µl

Mix and incubate for 5 min. at 37°C. Measure the absorbance of sample and standard against reagent blank.

Calculation

Cholesterol Conc. (mg/dL) = Absorbance of sample /Absorbance of standard × 200

= _____ mg/dl.

(Where **200 mg/dL** is the assigned value of the standard.)

Precaution

- To avoid contamination, use clean laboratory equipment's.
- Avoid direct exposure of working reagent to light.

Normal Range

It is recommended that each laboratory establish its own reference values. The following values may be used as guideline.

Normal serum/plasma cholesterol level is 150-250 mg/dl.

Clinical Significance

- Cholesterol is the main lipid found in the blood, bile and brain tissues.
- It is also one of the most important steroids of the body and is a precursor of many steroid hormones.
- Two thirds of cholesterol present in the blood is esterified. The liver metabolizes cholesterol, and it is transported in the blood stream by lipoproteins.
- Increased levels are found in hypercholesterolemia, hyperlipidemia, hypothyroidism, uncontrolled diabetes, nephrotic syndrome and cirrhosis.

- Decreased levels are found in malabsorption, malnutrition, hyperthyroidism, anemia and liver diseases.

EMULSIFICATION TEST

OBJECT: To demonstrate the process of emulsification of lipids in a given sample.

THEORY: Emulsion is a colloidal dispersion in which both the disperse phase and dispersion medium are liquids but are not miscible.

PRINCIPLE: When oil and water, which are not miscible, are shaken together, the oil is broken up into very tiny droplets which are dispersed in water, this is known as an oil in water emulsion. The water molecules due to their high surface tension have a tendency to come together and form a separate layer. That is why oil and water emulsion is unstable. In the presence of emulsifying

agent like bile salt, NaCO; and soaps, lipid molecule are emulsified and broken down into small droplets.

REAGENTS: Bile salts solution (5%)

PROCEDURE:

1. Take 2 test tubes and mark them A and B

2. In test tube A, take 1ml of given sample of oil and add 1ml of water.

3. In test tube B, take 1ml of given sample of oil and add 1ml of bile salt solution.

4. Shake them well and allow stand

OBSERVATION:

RESULT: